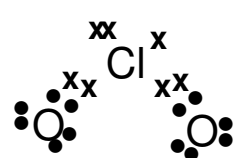
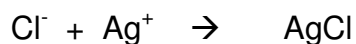
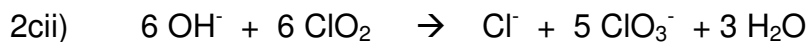


**Yishun Junior College 2008 Preliminary examination
H2 Chemistry P2**

Qn No	Answer	Marks
1ai)	Nitroglycerine : Gaseous products $\begin{array}{ccc} 4 & : & 29 \\ 4/29 & : & 1 \end{array}$ $(101000)(83 \times 10^{-3}) = n(8.31)(773)$ $n = 1.305$ $n_{\text{nitroglycerine}} = 4/29 (1.305)$ $= 0.1800 \text{ mol}$ $\text{Mass of nitroglycerine} = (0.1800)(227)$ $= 40.86 \text{ g}$	1
1aii)	A high volume of gaseous products formed, resulting in high pressure leading to an explosion or high pressure exerted by gaseous products.	1
1b)	N ₂ (g), smallest electron cloud or molecule. Hence, volume of molecule and induced dipoles attraction are negligible.	1 + 1
1c)	$PV = 5.82 \times 10^4 / 112$ $= 519.6 \text{ kPa}$ $= 520 \text{ kPa}$	1 1
1d)	O ₂ ; CO ₂	½ X 1 = 1 Total 8
2a)	i) Si or Al ii) Al iii) Cl iv) Mg v) Si vi) C	1/2 1/2 1/2 1/2 1/2 1/2
2bi)	 Shape : Bent	1 1
2bii)	$2 \text{H}^+ + \text{ClO}_3^- + \text{e}^- \rightarrow \text{ClO}_2 + \text{H}_2\text{O}$ $\text{H}_2\text{O}_2 \rightarrow \text{O}_2 + 2 \text{H}^+ + 2 \text{e}^-$ $2 \text{H}^+ + 2 \text{ClO}_3^- + \text{H}_2\text{O}_2 \rightarrow 2 \text{ClO}_2 + \text{O}_2 + 2 \text{H}_2\text{O}$	1
2ci)	Element undergoes reduction and oxidation at the same time.	1



$$n_{\text{Ag}^+} = \frac{20}{1000} \times 0.05 = 1.00 \times 10^{-3} \text{ mol}$$

1/2

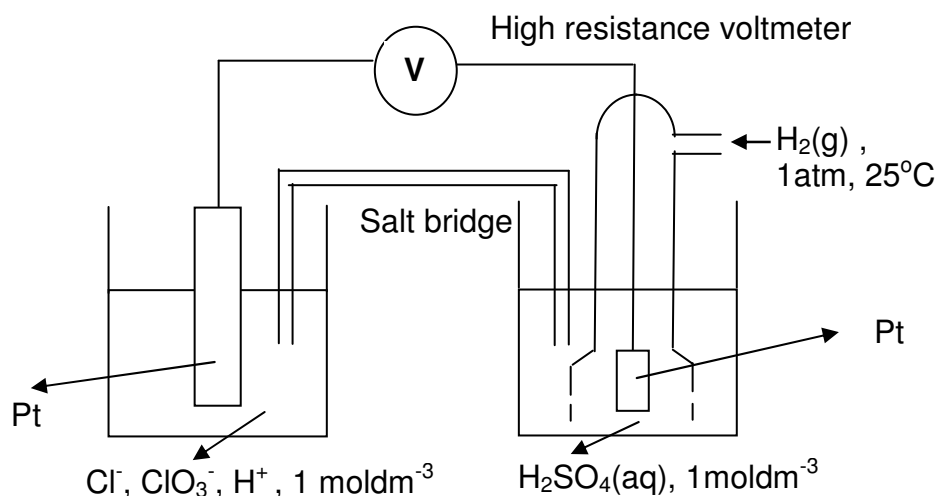
$$n_{\text{ClO}_2} = 6.00 \times 10^{-3} \text{ mol}$$

1/2

$$V_{\text{ClO}_2} = (6 \times 10^{-3}) (24) = 0.144 \text{ dm}^3$$

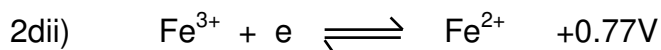
1

2di)



Set-up =1

Labels=2
 $\text{H}_2(\text{g})$,
 1 atm, 25°C
 $\text{H}_2\text{SO}_4(\text{aq})$,
 1 moldm⁻³
 Cl^- , ClO_3^- ,
 H^+ , 1
 moldm⁻³
 Salt bridge
 or
 voltmeter
 or Pt
 1/2 each



$$E^\ominus_{\text{cell}} = 1.45 - 0.77 = +0.68 \text{ V}$$

1

$E^\ominus_{\text{cell}}$ is positive. $\therefore$ Feasible reaction

1

2diii) Both Fe^{3+} and Fe^{2+} will be precipitated. Therefore, reduction potential will not be affected.

1/2

$[\text{ClO}_3^-]$ increases as H^+ is neutralised by added OH^- .

1/2

$E^\ominus_{\text{red}}$ become less positive.

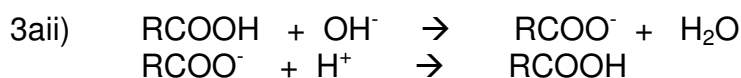
1/2

Hence, $E^\ominus_{\text{cell}}$ become less positive.

1/2

3ai) $\text{pH} = \text{pKa} = 3.13$

1



1

1

3b) $\text{pH} = \text{pK}_a + \lg [\text{salt}] / [\text{acid}]$

$$= -\lg (7.4 \times 10^{-4}) + \lg \frac{[20 \times 0.1 / 50]}{[10 \times 0.1 / 50]}$$

1

$$= 3.13 + 0.301$$

$$= 3.43$$

1

3c) $K_b = x^2 / 0.05$

$$x = 8.367 \times 10^{-7}$$

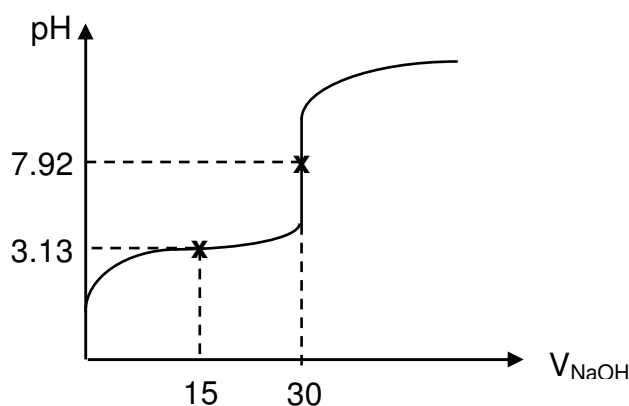
$$\text{pOH} = 6.07$$

$$\text{pH} = 7.92$$

1

1

3d)



$$\text{Vol} = \frac{1}{2} \times 2$$

$$= 1$$

$$\text{pH} = \frac{1}{2} \times 2$$

$$= 1$$

curve=1

4a) +2

1

Co-ordination number = 6

1

4b) Oxidising agent

1

4c) The d-orbitals split into 2 different energy levels. The promotion of the d electrons from the lower energy level to higher energy level absorbs wavelength in the visible region. The complementary colour not absorbed is transmitted, giving rise to colour of complex.

1

1

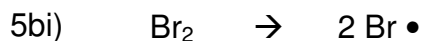
1

5a) Step I : electrophilic addition

1

Step III : free radical substitution

1



1

5bii) HBr (g)

1

5biii) $\text{sp}^3, 109^\circ$

1+ 1



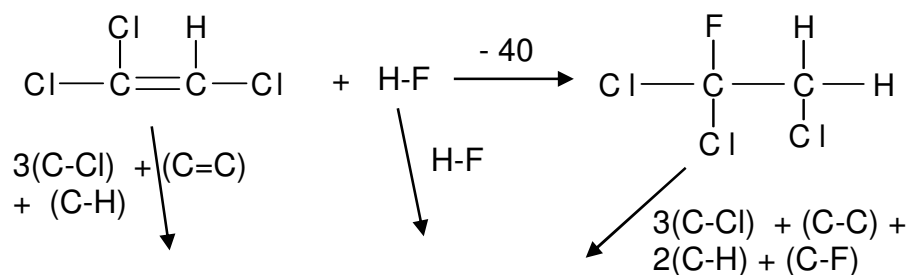
Mass of fluothane = $10 / 118.5 \times (197.4)$
 = 16.7 g

1

% yield = $9 / 16.7 \times 100\%$
 = 54.0%

1

5d)



Need not
show cycle

$610 + 410 + 562 = -40 + 350 + 2(410) + \text{BE}(\text{C}-\text{F})$

1

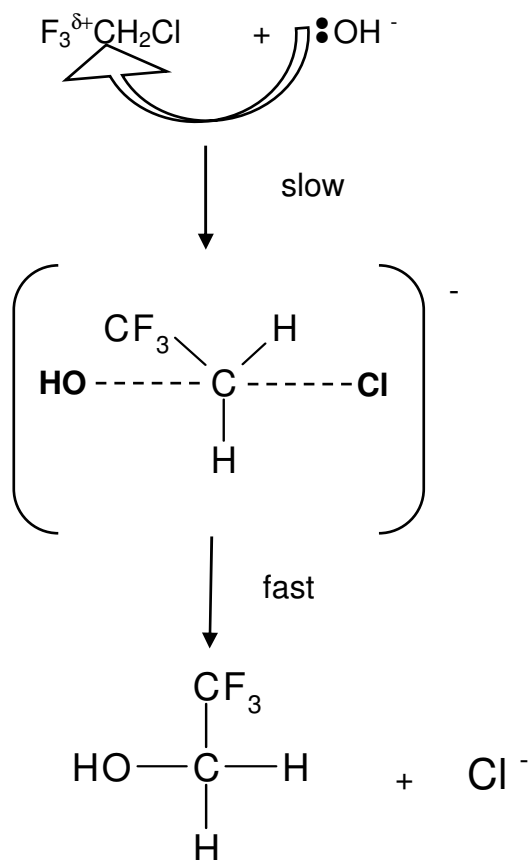
$\text{BE}(\text{C}-\text{F}) = 452 \text{ kJmol}^{-1}$

1

5ei) Aqueous NaOH

1

ii)



indicate
Slow $\frac{1}{2}$
Fast $\frac{1}{2}$

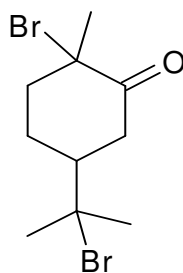
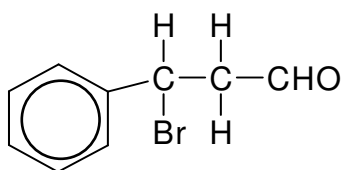
Reaction
of slow
step and
fast step
 $\frac{1}{2}$ each

6a) D, violet or purple solution.

1

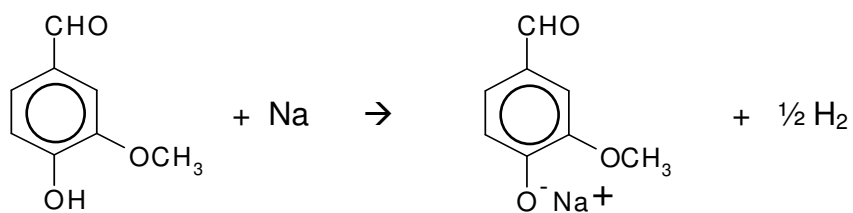
6b)

1 + 1



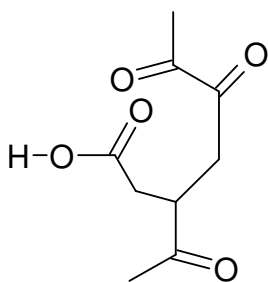
6c)

1



6d)

2



6e) $\text{H}_2(\text{g}) / \text{Ni}$

1